PECH GROUP HOLDINGS LTD

Technology & Infrastructure Enablers for Africa

PECH GROUP HOLDINGS

PECH GROUP HOLDINGS LTD

AI Architecture Guide — Technical Blueprint

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1. ARCHITECTURE OVERVIEW

System Layers

```
┌──────────────────────────────────────────────────────────────┐
│ LAYER 5: USER INTERFACES                                     │
│ Web Apps │ Mobile Apps │ WhatsApp Bot │ API Consumers │ IoT  │
├──────────────────────────────────────────────────────────────┤
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├──────────────────────────────────────────────────────────────┤
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```

```
│ Identity │ Commerce │ Solar │ Fintech │ Logistics │ IoT │ ...  │
├──────────────────────────────────────────────────────────────┤
│ LAYER 2: AI CORE                                             │
│ LLM Serving │ RAG │ Vision │ Speech │ ML │ Embeddings │ Agents │
├──────────────────────────────────────────────────────────────┤
│ LAYER 1: DATA & INFRASTRUCTURE                               │
│ PostgreSQL │ Qdrant │ Redis │ NATS/Kafka │ Blob Storage       │
└──────────────────────────────────────────────────────────────┘
```

Design Principles

1. **AI-native** — AI is not a bolt-on; every service talks to the AI Core
2. **Self-hosted first** — all models and data on PECH infrastructure
3. **Cloud burst** — overflow to cloud GPU when local capacity exhausted
4. **API-first** — all services expose REST/gRPC APIs via APISIX
5. **Event-driven** — services communicate via NATS/Kafka, not direct calls
6. **Multi-tenant** — each vertical's users are isolated (Keycloak realms)

2. AI CORE DESIGN

The AI Core is the central intelligence layer. All business services call it via internal APIs.

AI Core Components

```
┌──────────────────────────────────────────────┐
│                  AI CORE                     │
│                                              │
│  ┌──────────────┐  ┌──────────────────────┐  │
│  │ Model Router │──│ Ollama / vLLM        │  │
│  │ (APISIX)     │  │ (LLM Serving)        │  │
│  └──────────────┘  └──────────────────────┘  │
│         │                                    │
│  ┌──────▼───────┐  ┌──────────────────────┐  │
│  │ RAG Engine   │──│ Qdrant (Vectors)     │  │
│  │ (LangChain)  │  │ + Embeddings         │  │
│  └──────────────┘  └──────────────────────┘  │
│                                              │
│  ┌──────────────┐  ┌──────────────────────┐  │
│  │ Agent System │  │ Specialized Models   │  │
│  │ (LangGraph)  │  │ STT │ TTS │ OCR │ CV │  │
│  └──────────────┘  └──────────────────────┘  │
│                                              │
│  ┌──────────────────────────────────────┐    │
│  │ ML Pipeline (XGBoost, Prophet, LightFM) │  │
│  └──────────────────────────────────────┘    │
└──────────────────────────────────────────────┘
```

Model Serving Architecture

Phase 1: Ollama (Simple)

```
Client → APISIX → Ollama API (port 11434)
                  ├── qwen2.5:7b (loaded)
```

```
                    ├── qwen2.5-coder:7b (loaded on demand)
                    └── nomic-embed-text (loaded)
```

- **Pros:** Single binary, easy model management, pull & run
- **Cons:** No request batching, lower throughput, single-queue
- **Capacity:** ~5-10 concurrent users

Phase 2: vLLM (Production)

```
Client → APISIX → Load Balancer
                    ├── vLLM Instance 1 (Qwen2.5-14B, GPU 0)
                    ├── vLLM Instance 2 (Qwen2.5-Coder-14B, GPU 0)
                    └── vLLM Instance 3 (Qwen2.5-32B, GPU 1)
```

- **Pros:** Continuous batching, PagedAttention, OpenAI-compatible API
- **Cons:** More complex setup
- **Capacity:** ~50-100 concurrent users per GPU

Model Routing Logic

APISIX routes requests to the appropriate model based on the request type:

Request Type	Model	Priority
General chat	Qwen2.5-7B/14B	Default
Code generation	Qwen2.5-Coder-7B/14B	Detect code context
Complex reasoning	Qwen2.5-32B	Premium tier or flagged queries
Image analysis	Qwen2.5-VL-7B	Image attached
Voice input	Faster-Whisper → LLM	Audio attached
Embeddings	all-MiniLM / Nomic	Internal RAG calls

3. RAG PIPELINE ARCHITECTURE

RAG (Retrieval-Augmented Generation) is how PECH's AI answers questions using business data without fine-tuning.

Pipeline Flow

```
┌──────────────────────────────────────────────────────────┐
│ DATA INGESTION (Apache Airflow — scheduled daily)        │
│                                                          │
│  Sources:                                                │
│  ├── Product catalog (Medusa) ──────┐                    │
│  ├── Support tickets (Chatwoot) ────┤                    │
│  ├── Solar manuals (PDF/docs) ──────┤                    │
│  ├── Financial data (ERPNext) ──────┤                    │
│  ├── IoT device docs (ThingsBoard) ─┤                    │
│  ├── Knowledge base (Strapi) ───────┤                    │
│  └── HR policies (internal docs) ───┘                    │
│                                     │                    │
│                              ┌──────▼──────┐             │
│                              │ Chunking    │             │
│                              │ (LangChain) │             │
│                              │ 512 tokens  │             │
│                              └─────────────┘             │
```

```
│                                   │
│                                   │
│                                   │
```

Services organized by domain. Each service is a Docker container with its own database schema in PostgreSQL.

Domain 1: Identity & Access (3 services)

#	Service	Tech	Purpose
1	auth-service	Keycloak	SSO, OAuth2, OIDC, social login
2	user-profile-service	Node.js	User profiles, preferences, KYC
3	rbac-service	Node.js	Role-based access control, permissions

Domain 2: Core User Platform (6 services)

#	Service	Tech	Purpose
4	notification-service	Node.js	Email, SMS, push, WhatsApp notifications
5	search-service	Python	Unified search across all verticals
6	file-service	Node.js	File upload, storage, CDN
7	billing-service	Node.js	Subscription billing, invoicing
8	analytics-service	Python	User analytics, event tracking
9	feedback-service	Node.js	Reviews, ratings, NPS

Domain 3: Marketplace & Commerce (10 services)

#	Service	Tech	Purpose
10	product-catalog-service	Medusa	Product listings, categories, attributes
11	vendor-service	Node.js	Vendor onboarding, verification, management
12	order-service	Medusa	Order lifecycle, tracking
13	cart-service	Medusa	Shopping cart, wishlist
14	payment-service	Node.js	Paystack/Flutterwave integration
15	pricing-service	Python	Dynamic pricing, discounts (AI-powered)
16	shipping-service	Node.js	Shipping rates, tracking integration
17	review-service	Node.js	Product reviews, seller ratings
18	recommendation-service	Python	LightFM product recommendations
19	image-service	Python	Background removal (rembg), enhancement

Domain 4: Solar & Energy (8 services)

#	Service	Tech	Purpose
20	solar-product-service	Node.js	Solar panels, inverters, batteries catalog
21	solar-calculator-service	Python	System sizing, ROI calculator
22	installer-service	Node.js	Installer marketplace, certification
23	installation-service	Node.js	Job scheduling, tracking, completion
24	warranty-service	Node.js	Warranty registration, claims
25	energy-monitor-service	Python	ThingsBoard integration, energy analytics
26	maintenance-service	Python	Predictive maintenance (AI-powered)

27	solar-inspection-service	Python	Panel defect detection (RT-DETR)

Domain 5: Fintech & Payments (8 services)

#	Service	Tech	Purpose
28	wallet-service	Fineract	Digital wallet, balance, transfers
29	agent-banking-service	Fineract	Cash-in/cash-out agent network
30	loan-service	Fineract	Micro-loans for solar equipment
31	savings-service	Fineract	Savings accounts, interest
32	bill-payment-service	Node.js	Utility bills, airtime, cable
33	kyc-service	Python	KYC verification (OCR + face match)
34	fraud-detection-service	Python	XGBoost + Isolation Forest
35	transaction-ledger-service	Fineract	Transaction history, statements

Domain 6: Logistics & Delivery (7 services)

#	Service	Tech	Purpose
36	route-optimization-service	Python/OSRM	Delivery route planning
37	fleet-service	Node.js	Vehicle management, GPS tracking
38	dispatch-service	Node.js	Order-to-driver assignment
39	tracking-service	Node.js	Real-time delivery tracking
40	hub-service	Node.js	Hub/warehouse management
41	last-mile-service	Node.js	Last-mile delivery management
42	delivery-pricing-service	Python	Delivery cost calculation

Domain 7: Real Estate (5 services)

#	Service	Tech	Purpose
43	property-listing-service	Node.js	Property listings, search
44	property-valuation-service	Python	AI property valuation
45	virtual-tour-service	Node.js	360° tours, image galleries
46	agent-matching-service	Python	LightFM agent-client matching
47	rental-management-service	Node.js	Lease management, payments

Domain 8: Procurement & Tenders (5 services)

#	Service	Tech	Purpose
48	tender-service	Python	OpenProcurement integration
49	bid-management-service	Node.js	Bid submission, evaluation
50	supplier-service	Node.js	Supplier registry, verification
51	contract-service	Node.js	Contract management, milestones
52	procurement-analytics-service	Python	Spend analysis, savings

Domain 9: IoT & Monitoring (5 services)

#	Service	Tech	Purpose
53	device-management-service	ThingsBoard	Device registration, provisioning
54	telemetry-service	ThingsBoard	Real-time data collection
55	alert-service	Python	Rule-based and AI-powered alerts
56	dashboard-service	ThingsBoard	Customer-facing IoT dashboards
57	firmware-update-service	Node.js	OTA firmware updates

Domain 10: Customer Support (5 services)

#	Service	Tech	Purpose
58	chatbot-service	Python	AI chatbot (Qwen2.5 + RAG)
59	ticket-service	Chatwoot	Support ticket management
60	knowledge-base-service	Python	RAG-powered knowledge base
61	sentiment-service	Python	DistilBERT sentiment analysis
62	escalation-service	Node.js	Auto-escalation rules

Domain 11: AI & Data Platform (10 services)

#	Service	Tech	Purpose
63	llm-gateway-service	Python	Route to appropriate LLM
64	embedding-service	Python	Generate embeddings for RAG
65	rag-service	Python	RAG pipeline orchestration
66	stt-service	Python	Speech-to-text (Faster-Whisper)
67	tts-service	Python	Text-to-speech (Piper)
68	ocr-service	Python	Document OCR (PaddleOCR)
69	vision-service	Python	Image analysis (Qwen-VL)
70	ml-prediction-service	Python	XGBoost/LightGBM predictions
71	training-pipeline-service	Python	Model fine-tuning orchestration
72	data-ingestion-service	Python	ETL pipelines (Airflow)

Domain 12: Accounting & Business (6 services)

#	Service	Tech	Purpose
73	accounting-service	ERPNext/Frappe	GL, AP, AR, journal entries
74	tax-service	Python	FIRS compliance, VAT, WHT
75	audit-trail-service	Python	Transaction audit logging
76	payroll-service	ERPNext	Employee payroll
77	hr-service	Python	HR management, recruitment AI
78	reporting-service	Python	Apache Superset integration

5. API GATEWAY DESIGN

Apache APISIX Configuration

```
Internet → DNS → Nginx (TLS) → APISIX → Microservices
                                  │
                                  ├── /api/v1/auth/*     → Keycloak
                                  ├── /api/v1/products/* → product-catalog-service
                                  ├── /api/v1/ai/chat    → llm-gateway-service
                                  ├── /api/v1/ai/embed   → embedding-service
                                  ├── /api/v1/ai/stt     → stt-service
                                  ├── /api/v1/ai/ocr     → ocr-service
                                  ├── /api/v1/solar/*    → solar-*-service
                                  ├── /api/v1/fintech/*  → wallet/agent/loan-service
                                  ├── /api/v1/logistics/* → route/fleet/dispatch-service
                                  └── /api/v1/iot/*      → ThingsBoard
```

Rate Limiting by Tier

Tier	AI Requests/min	General API/min	Monthly Limit
Free	10	60	100 AI queries
Basic (₦2,000/mo)	30	300	1,000 AI queries
Pro (₦5,000/mo)	100	1,000	10,000 AI queries
Enterprise	Custom	Custom	Unlimited

6. AI AGENT FRAMEWORK

LangGraph for Multi-Step AI Tasks

LangGraph enables AI agents that can plan, execute, and iterate on complex tasks:

Example: Solar System Design Agent

```
User: "Design a solar system for my 3-bedroom house in Lekki"
  │
  ├── Step 1: Estimate load (AI calculates from house type)
  │     → Estimated load: 5kW peak, 25kWh/day
  │
  ├── Step 2: Query product catalog (RAG → Medusa products)
  │     → Retrieved: 5× 400W panels, 5kVA inverter, 200Ah LiFePO4
  │
  ├── Step 3: Calculate system (solar-calculator-service)
  │     → System: 2kWp, ~8kWh/day solar, ₦2.1M installed
  │
  ├── Step 4: Find installers (installer-service + LightFM matching)
  │     → Top 3 certified installers in Lekki area
  │
  └── Step 5: Generate proposal (LLM writes formatted proposal)
        → PDF with specs, pricing, installer options, ROI analysis
```

7. DATA FLOW PER VERTICAL

Solar Vertical — Complete Flow

```
Customer → Web/Mobile App
  │
  ├── Browse products → product-catalog-service → Medusa
  ├── AI chat about solar → chatbot-service → RAG → Qwen2.5
  ├── System design → solar-calculator-service → AI agent
  ├── Order panels → order-service → payment-service → Paystack
  ├── Schedule install → installation-service → installer-service
  ├── Monitor system → energy-monitor-service → ThingsBoard → IoT
  ├── Detect faults → solar-inspection-service → RT-DETR
  ├── Maintenance alert → maintenance-service → notification-service
  └── Support request → chatbot-service → ticket-service → Chatwoot
```

Fintech Vertical — Complete Flow

```
Customer/Agent → Mobile App
  │
  ├── Create wallet → wallet-service → Fineract
  ├── KYC verification → kyc-service → PaddleOCR + face match
  ├── Deposit (agent) → agent-banking-service → Fineract
  ├── Transfer → wallet-service → fraud-detection-service → XGBoost
  ├── Apply for loan → loan-service → credit scoring (XGBoost)
  ├── Pay bills → bill-payment-service → Paystack
  ├── View statement → transaction-ledger-service → Fineract
  └── Support → chatbot-service → RAG → Chatwoot
```

8. EVENT STREAMING ARCHITECTURE

NATS (Phase 1) / Kafka (Phase 3) Event Topics

Topic	Publisher	Subscribers	Events
`order.created`	order-service	payment, shipping, notification	New order placed
`payment.completed`	payment-service	order, notification, accounting	Payment confirmed
`product.updated`	product-catalog	search, recommendation, embedding	Product changed
`iot.telemetry`	telemetry-service	alert, dashboard, analytics	Device data
`ai.embedding.request`	data-ingestion	embedding-service	New data to embed
`support.ticket.created`	ticket-service	sentiment, escalation	New support ticket
`fraud.alert`	fraud-detection	notification, compliance	Suspicious transaction
`solar.fault.detected`	inspection-service	maintenance, notification	Panel defect found

9. USER-FACING AI FEATURES

Feature Matrix by Subscription Tier

Feature	Free	Basic	Pro	Enterprise
AI Chat Assistant	100 queries/mo	1,000/mo	10,000/mo	Unlimited
Product Recommendations	Basic	Personalized	Advanced	Custom model

Image Background Removal	5/mo	50/mo	500/mo	Unlimited
Voice-to-Edit	—	10/mo	100/mo	Unlimited
AI Product Descriptions	—	20/mo	200/mo	Unlimited
Dynamic Pricing Insights	—	—	✓	✓
Solar System Design AI	Basic	Full	Full + custom	White-label
Invoice OCR	5/mo	50/mo	500/mo	Unlimited
AI Financial Reports	—	—	✓	✓

Image Background Removal (rembg)

```
User uploads product photo
  → image-service receives image
  → rembg (U²-Net) removes background
  → Real-ESRGAN upscales if low-res
  → Return clean product image on white/transparent background
  → User can choose template backgrounds
```

Voice-to-Edit

```
User speaks edit command (e.g., "Change the price to 15,000 naira")
  → stt-service (Faster-Whisper) → text: "Change the price to 15,000 naira"
  → NLU parsing (Qwen2.5) → intent: update_price, value: 15000, currency: NGN
  → API call to product-catalog-service
  → Confirmation via tts-service (Piper TTS) → "Price updated to 15,000 naira"
```


10. ACCOUNTING, AUDITING & TAX AI

Subscription Product: PECH AI Accountant

A standalone AI-powered accounting tool that accountants and businesses subscribe to:

Tier	Price (₦/mo)	Features
Sole Trader	3,000	Invoice OCR, basic bookkeeping, VAT calculation
SME	8,000	Full GL, bank reconciliation, FIRS filing, payroll
Enterprise	25,000	Multi-entity, audit trail, custom reports, API access
Accounting Firm	50,000	Multi-client, team access, white-label reports

AI Features

Feature	Model/Tool	How It Works
Invoice OCR	PaddleOCR → Qwen2.5	Scan invoice → extract fields → create journal entry
Bank Reconciliation	XGBoost	Match bank transactions to GL entries (pattern learning)
Tax Computation	Rule engine + Qwen2.5	Calculate VAT (7.5%), WHT, CIT based on transaction type

Audit Anomaly Detection	Isolation Forest	Flag unusual transactions, duplicate payments, round amounts
Financial Forecasting	Prophet	Revenue/expense forecasting from historical data
Natural Language Queries	Qwen2.5 + RAG	"What were our top 5 expenses last quarter?"
Auto-Categorization	DistilBERT	Classify transactions into chart of accounts categories
FIRS Filing Assistant	Qwen2.5 + templates	Generate FIRS-compliant tax returns from accounting data

11. HR & RECRUITMENT AI

AI Features for HR Module

Feature	Model/Tool	How It Works
Resume Parsing	PaddleOCR + Qwen2.5	Extract structured data from CV (PDF/image)
Candidate Scoring	XGBoost	Score candidates against job requirements
Interview Q&A Generation	Qwen2.5 + RAG	Generate role-specific interview questions from job handbook
Skill Gap Analysis	Qwen2.5	Compare candidate skills to role requirements
Offer Letter Generation	Qwen2.5 + templates	Auto-generate offer letters with correct salary/benefits
Onboarding Checklist	Rule engine	Auto-assign tasks based on role and department
Performance Review AI	Qwen2.5	Summarize peer feedback, suggest development areas

12. SECURITY & NDPA COMPLIANCE

Nigeria Data Protection Act (NDPA) 2023

Requirement	PECH Implementation
Data minimization	Collect only necessary data; anonymize where possible
Consent management	Keycloak consent flows; explicit opt-in for AI features
Data residency	All primary data stored on Nigeria-based servers
Right to erasure	User deletion API in user-profile-service
Data breach notification	72-hour notification via alert-service
DPO appointment	Required — assign Data Protection Officer
DPIA	Data Protection Impact Assessment for AI features

Security Architecture

Layer	Implementation
Network	pfSense firewall, VPN for remote access, private VLAN for servers
Application	APISIX rate limiting, JWT auth, CORS, input validation
Data	PostgreSQL row-level security, encryption at rest (LUKS), TLS in transit
AI	Prompt injection filtering, output sanitization, no PII in embeddings
Access	Keycloak RBAC, MFA for admin, audit logging
Monitoring	Fail2ban, OSSEC, Prometheus alerting

13. DEPLOYMENT ARCHITECTURE

Phase 1: Docker Compose

```
┌─────────────────────────────────────┐
│   TRX50 WORKSTATION (Phase 1)       │
│                                     │
│  Docker Compose                     │
│  ├── ollama (GPU)                   │
│  ├── qdrant                         │
│  ├── postgres                       │
│  ├── redis                          │
│  ├── faster-whisper (GPU)           │
│  ├── paddleocr                      │
│  ├── keycloak                       │
│  ├── medusa                         │
│  ├── chatwoot                       │
│  ├── apisix                         │
│  ├── prometheus + grafana           │
│  └── application services           │
│                                     │
│  Portainer (management UI)          │
└─────────────────────────────────────┘
```

Phase 2: K3s (Lightweight Kubernetes)

```
┌─────────────────────────────────────┐
│             K3s CLUSTER             │
│                                     │
│  Node 1 (TRX50 Workstation — upgraded)  │
│  ├── AI workloads (Ollama/vLLM, Whisper)  │
│  ├── Qdrant                         │
│  └── GPU-dependent services         │
│                                     │
│  Node 2 (Application server)        │
│  ├── Business microservices         │
│  ├── Keycloak, Medusa, ERPNext      │
│  └── APISIX, monitoring             │
│                                     │
│  Node 3 (Database server)           │
│  ├── PostgreSQL (primary)           │
│  ├── Redis                          │
│  └── NATS                           │
└─────────────────────────────────────┘
```


14. NIGERIA-SPECIFIC CONSIDERATIONS

Challenge	Impact	Architecture Solution
Power outages	Services go offline	UPS + solar + auto-restart Docker containers
Slow internet	High latency for users	Self-hosted everything; CDN for static assets

Slow internet	High latency for users	Self-hosted everything; CDN for static assets
Bandwidth costs	Expensive data	Compress API responses; optimize image sizes
Mobile-first users	70%+ mobile traffic	Mobile-optimized APIs; progressive web app
USSD banking	Feature phones need access	USSD gateway for fintech services
Multiple languages	Yoruba, Hausa, Igbo, Pidgin	Whisper for voice; Argos for text translation
Cash-heavy economy	Digital payments adoption	Agent banking network (Fineract)
Low smartphone storage	Users can't install large apps	PWA (progressive web app) instead of native

15. AI TEAM STRUCTURE

Organization Chart

```
┌──────────────────────────────────┐
│      CTO / Managing Director     │
└──────────────────────────────────┘
                │
    ┌───────────┼───────────┐
    │           │           │
┌───▼───┐   ┌───▼───┐   ┌───▼───┐
│ AI &  │   │Product│   │ Ops & │
│ Eng.  │   │ & Biz │   │ Infra │
└───────┘   └───────┘   └───────┘
    │           │           │
    ├── AI/ML Engineer (×2)        ₦600K-₦1M/mo
    ├── AI Data Engineer            ₦400K-₦600K/mo
    ├── MLOps Engineer              ₦500K-₦800K/mo
    ├── API/Platform Engineer       ₦450K-₦700K/mo
    │           │
    │           ├── AI Product Manager ₦500K-₦800K/mo
    │           ├── Developer Advocate ₦400K-₦600K/mo
    │           ├── Technical Writer   ₦250K-₦400K/mo
    │           │
    │           │           ├── Installer Training Mgr  ₦350K-₦500K/mo
    │           │           ├── Fintech Ops / Agent Mgr  ₦400K-₦600K/mo
    │
    └── AI/ML Intern (×2)          ₦80K-₦120K/mo
```

Monthly Team Cost

Role	Count	Monthly (₦)	Annual (₦)
AI/ML Engineers	2	1,200,000-2,000,000	14,400,000-24,000,000
AI Data Engineer	1	400,000-600,000	4,800,000-7,200,000
MLOps Engineer	1	500,000-800,000	6,000,000-9,600,000
API/Platform Engineer	1	450,000-700,000	5,400,000-8,400,000
AI Product Manager	1	500,000-800,000	6,000,000-9,600,000
Developer Advocate	1	400,000-600,000	4,800,000-7,200,000
Technical Writer	1	250,000-400,000	3,000,000-4,800,000
Installer Training Mgr	1	350,000-500,000	4,200,000-6,000,000
Fintech Ops Mgr	1	400,000-600,000	4,800,000-7,200,000
AI/ML Interns	2	160,000-240,000	1,920,000-2,880,000

Total AI Team	**12**	**₦4,610,000-₦7,240,000**	**₦55,320,000-₦86,880,000**

Hiring Timeline

Phase	Month	Hires
Phase 1	1-3	2× AI/ML Interns, 1× AI/ML Engineer
Phase 2	4-9	1× AI Data Engineer, 1× API/Platform Engineer, 1× Installer Training Mgr, 1× Fintech Ops Mgr, 1× Technical Writer
Phase 3	10-18	1× AI/ML Engineer #2, 1× MLOps Engineer, 1× AI Product Manager, 1× Developer Advocate

This document is confidential to PECH Group Holdings Ltd. Last updated: March 2026.

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